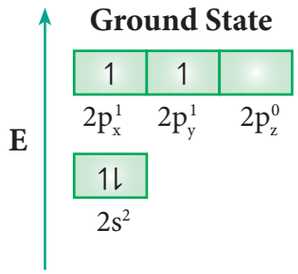
sp3 Hybridisation:

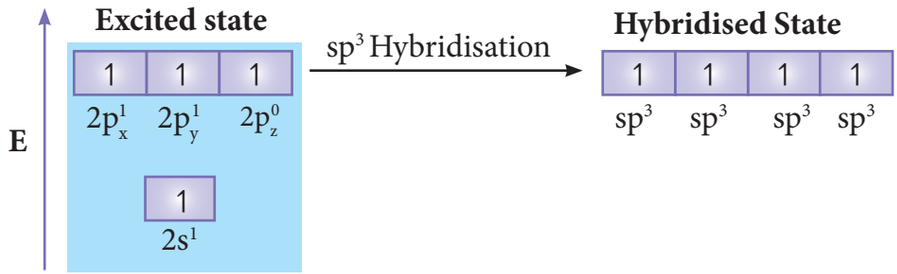
sp^3 hybridisation can be explained by considering methane as an example. In methane molecule the central carbon atom bound to four hydrogen atoms. The ground state valence shell electronic configuration of carbon is $[He]2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^0$.



In order to form four covalent bonds with the four hydrogen atoms, one of the paired electrons in the 2s orbital of carbon is promoted to its $2p_z$ orbital in the excite state. The one 2s orbital and the three 2p orbitals of carbon mixes to give four equivalent sp^3 hybridised orbitals. The angle between any two sp^3 hybridised orbitals is $109^0\ 28'$

Overlap with 1s orbitals of hydrogen:

The 1s orbitals of the four hydrogen atoms overlap linearly with the four sp^3 hybridised orbitals of carbon to form four C-H σ-bonds in the methane molecule, as shown below.



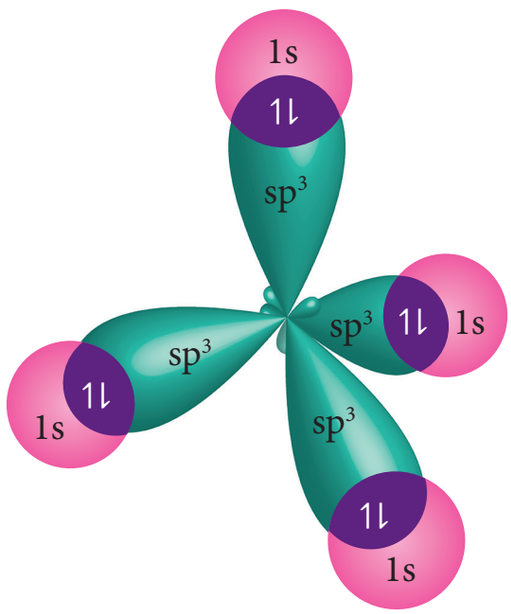


Fig 10.24 sp^3 Hybridisation : CH_4

11th Chemistry Chapter-10 Vol-2 EM.indd 92 09-01-2023 14:39:45